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COMPUTER NETWORKS

TEAM NETWORKS

Department of Computer Science and Engineering

Unit – 4 Network Layer and Internet Protocol

4.1 Overview of Network Layer

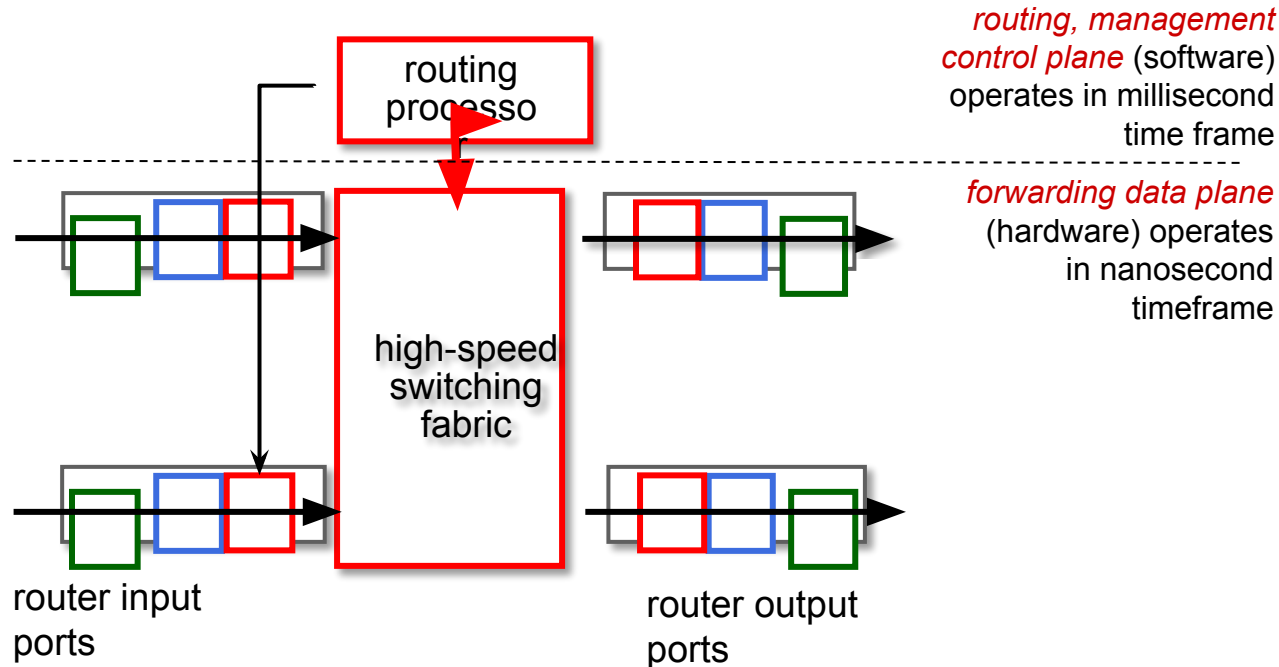
4.2 What's Inside a Router?

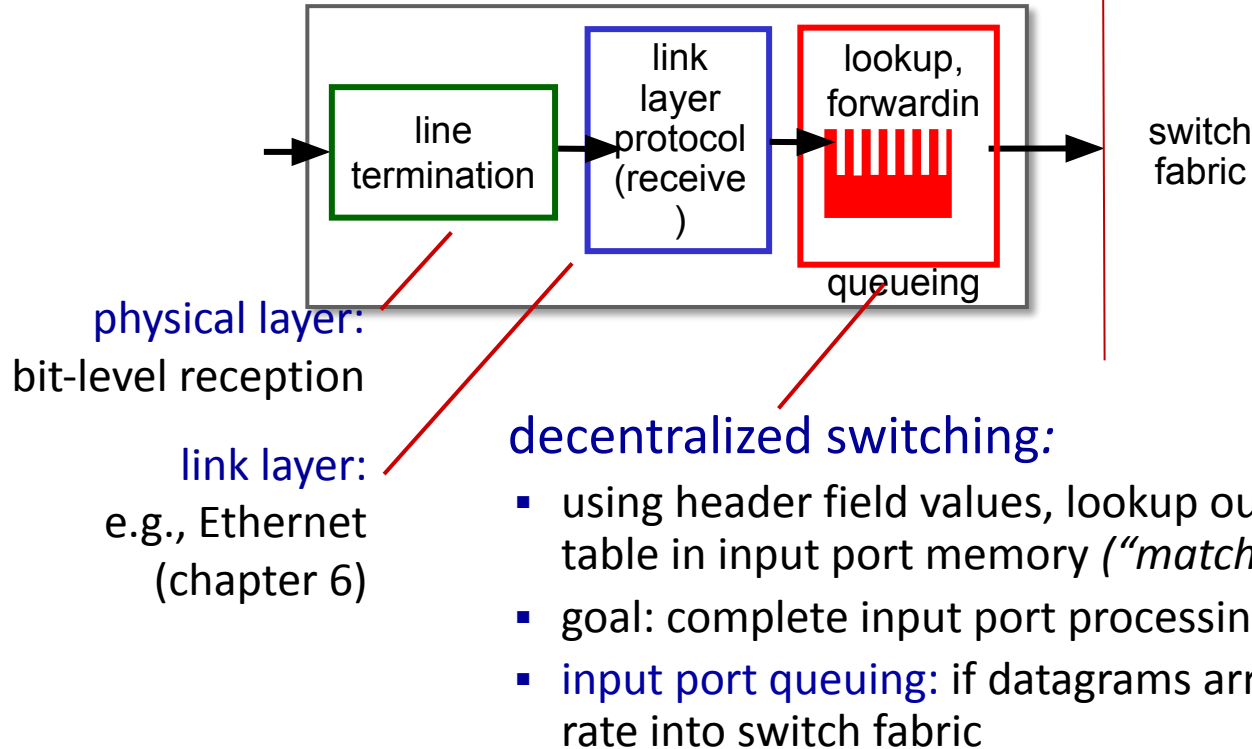
4.3 Switching

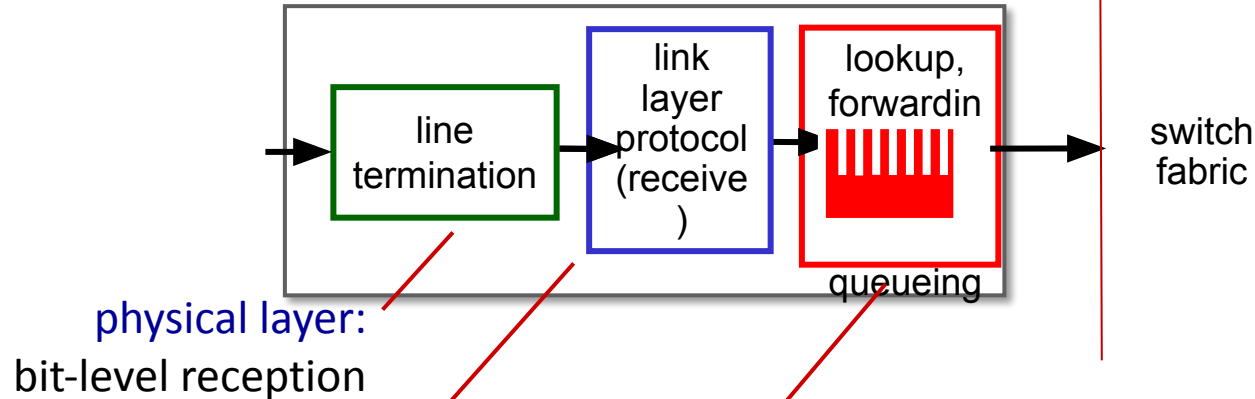
4.4 The Internet Protocol (IP)

- Datagram format
- Fragmentation
- IPv4 addressing
- NAT

high-level view of generic router architecture:







decentralized switching:

- using header field values, lookup output port using forwarding table in input port memory (*“match plus action”*)
- **destination-based forwarding:** forward based only on destination IP address (traditional)
- **generalized forwarding:** forward based on any set of header field values

forwarding table

Destination Address Range	Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010000 00000100 through 11001000 00010111 00010000 00000111	n 3
11001000 00010111 00011000 11111111	
11001000 00010111 00011001 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

Q: but what happens if ranges don't divide up so nicely?

longest prefix match

when looking for forwarding table entry for given destination address, use *longest* address prefix that matches destination address.

Destination Address Range	Link interface
11001000 00010111 00010 *** *****	0
11001000 00010111 00011000 *****	1
11001000 00010111 00011 *** *****	2
otherwise	3

examples:

11001000 00010111 00010110 10100001

which interface?

11001000 00010111 00011000 10101010

which interface?

longest prefix match

when looking for forwarding table entry for given destination address, use *longest* address prefix that matches destination address.

Destination Address Range	Link interface
11001000 00010111 00010*** *****	0
11001000 00010111 00011000 *****	1
11001000 00010111 00011*** *****	2
otherwise *	3

match

!

examples:

11001000 00010111 00010110 10100001	which interface?
11001000 00010111 00011000 10101010	which interface?

longest prefix match

when looking for forwarding table entry for given destination address, use *longest* address prefix that matches destination address.

Destination Address Range	Link
11001000 00010111 00010* *****	interface
11001000 00010111 0001*000 *****	0
11001000 00010111 0001*****	1
otherwise *	2

match!

examples:

11001000 00010111 00010110
 10100001 00010111 00011000
 10101010

3
which interface?

which interface?

longest prefix match

when looking for forwarding table entry for given destination address, use *longest* address prefix that matches destination address.

Destination Address Range	Link
11001000 00010111 00010* *****	interface
11001000 00010111 0001100 *****	0
11001000 00010111 00011* *****	1
otherwise *	2

match!

examples:

11001000 00010111 00010110
~~11001000 00010111 00010110~~
~~11001000 00010111 00010110~~
 10101010

3
which interface?

which interface?

- we'll see *why* longest prefix matching is used shortly, when we study addressing
- longest prefix matching: often performed using ternary content addressable memories (TCAMs)
 - *content addressable*: present address to TCAM: retrieve address in one clock cycle, regardless of table size
 - Cisco Catalyst: ~1M routing table entries in TCAM



THANK YOU

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